LICENSED READY-TO-USE GEMCITABINE BAGS CUT CARBON EMISSIONS VERSUS IN-HOUSE VIAL COMPOUNDING

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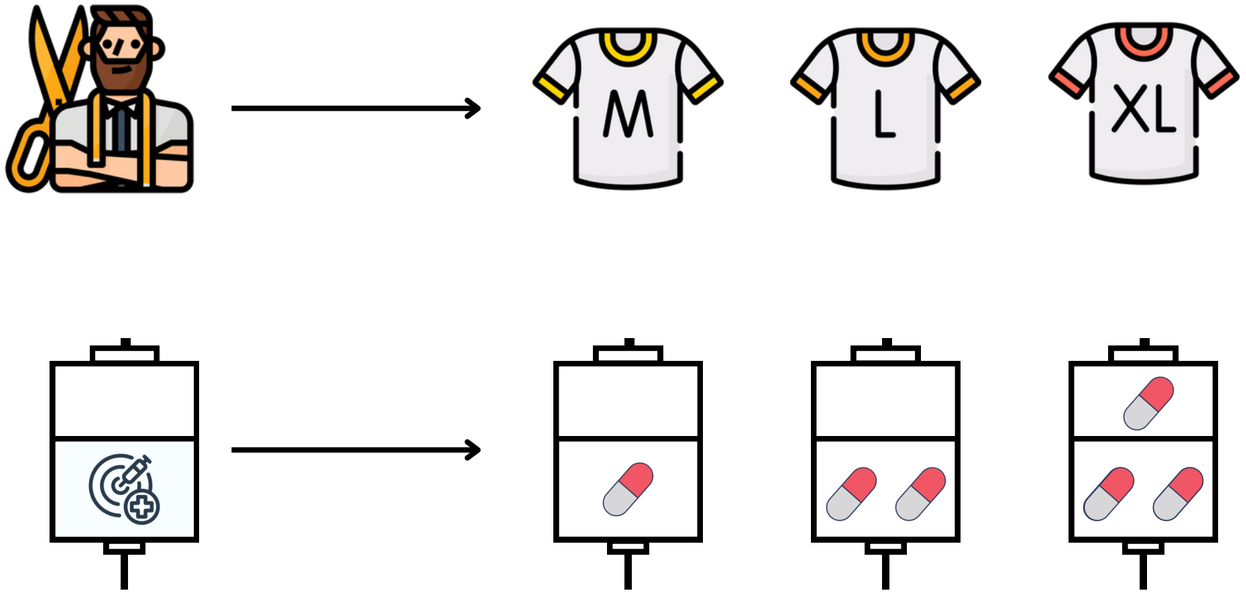
Disclosures

- Speaker contract with SunPharma
- Conference expenses covered by SunPharma
- Equity investment in a digital start-up for chemotherapy standardization

Introduction: A Common Issue

- Longer delay between diagnosis and initial management (+2 weeks over 5 years).
- Frequent saturation of day hospitals in France.
- On-demand compounding of infusion bags by hospital compounding units = limiting step.
- **Yet, 20–50% of preparations could be anticipated.**

Introduction: Solved through dose-banding?



- Optimizing chemotherapy production
- Securing the preparation of treatments
- Potential ecological impact to prove

Objective: To compare life cycle assessments

- ~~Pre reconstituted bags~~
- Commercial ready-to-use bags

Sources :
- INESS, standardisation et arrondissement des doses d'agents antinéoplasiques, revue systématique
- Chatelut et al : Dose banding as an alternative to body surface area-based dosing of chemo agents (BJC)
- Puisset et al, Investigating the clinical impact of dose-banding for weekly paclitaxel in patients with BC (BJCP)

Methodology

For each type of preparation (industrial vs reconstitution)

Conducting a life cycle assessment

ISO 14040

carebone
L'outil pour DÉCARBONER le soin

EcovaMed

Use of 2 tools (for weighting)

Including all process steps

Transport, manufacturing, consumables...

Results

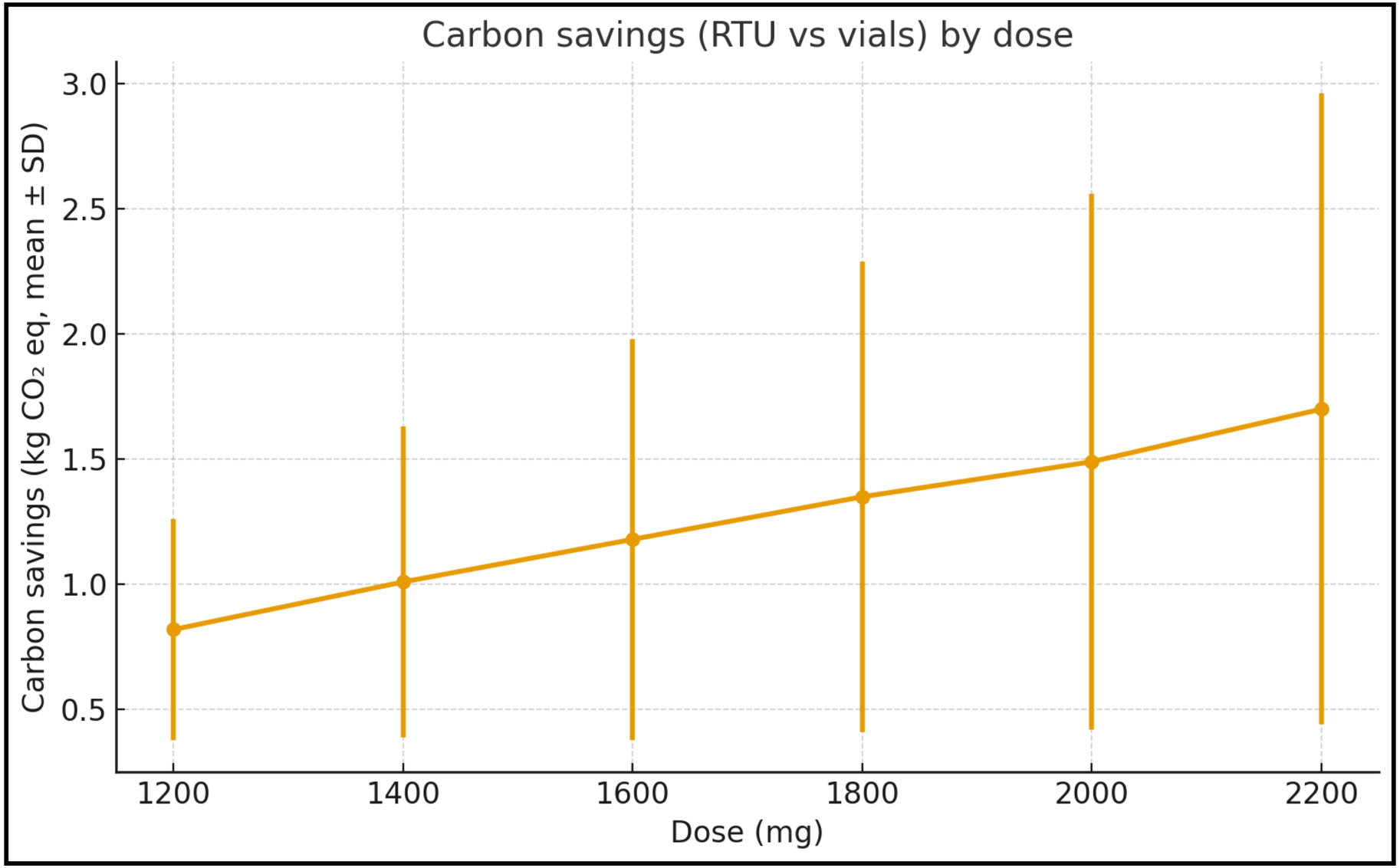
Dosage (mg)	Carbon savings (kg CO_2 eq, mean ± SD)	Relative savings vs traditional preparation
1200	0,82 ± 0,44	~ 45%
1400	1,01 ± 0,62	~ 45%
1600	1,18 ± 0,80	~ 50%
1800	1,35 ± 0,94	~ 50%
2000	1,49 ± 1,07	~ 55%
2200	1,70 ± 1,26	~ 55%

Glass → plastic (≈42%)

Avoided sterile devices (≈19%)

Reduced incinerated waste (≈14%)

Results


Carbon savings (RTU vs vials) by dose
Carbon savings (kg CO₂ eq, mean ± SD)
3.0
2.5
2.0
1.5
1.0
0.5
1200
1400
1600
1800
2000
2200
Dose (mg)

Discussion

What we already knew :

Dose banding is time saving.

Dose-banding has an economical impact.

What this study adds :

RTU Gemcitabine bags reduces the environmental impact of chemotherapy.

Is it also the case for dose-banding anticipated bags ?

Thanks for your attention